

University Course Registration System

Software Engineering Project

A web-based system that allows students to register, remove, and manage university courses efficiently.

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1 Problem Description

The current student course registration process in universities involves filling out registration forms manually, getting them signed by subject teachers, and then approved by advisors, deans, and accounts officers. After that, the forms are submitted to the administrative office.

This process is time-consuming and can lead to delays and errors. Students may face problems when looking for information about available courses, conflicts in schedules, and waiting for approvals. At the same time, administrative staff have to deal with a large amount of paperwork, which makes the process slower and less efficient.

To address these issues, an Online Student Course Registration System is proposed. This system will streamline the registration process, maintain accurate course information, reduce unnecessary paperwork, and monitor the registration process end-to-end. The system is designed to be simple, secure, and easy to use.

2 System Objectives

The system aims to:

- Enable students to register and remove courses online.
- Reduce the time required for course registration.
- Provide clear information about available courses.
- Allow administrators to manage courses, instructors, and schedules.
- Help avoid schedule conflicts.
- Reduce paperwork for administrative staff.
- Ensure the system is secure and easy to use.

3 Stakeholders

Stakeholder	Role	Interest in System
Students	Primary end-users	Enroll in courses, view schedules, check grades
Instructors	Course managers	Manage course sections, submit grades, view enrollment
Academic Advisors	Support staff	Monitor student progress, approve registrations
Registrar's Office	System administrators	Configure academic calendar, manage users and policies
IT Department	Technical owners	Deploy, maintain, and secure the application
University Management	Executive sponsors	Reporting, compliance, strategic planning

4 Work Breakdown Structure (WBS)

1. Project Planning

- 1.1. Select project topic
- 1.2. Define project scope
- 1.3. Assign team roles

2. System Analysis

- 2.1. Write project title
- 2.2. Prepare problem description
- 2.3. Define system objectives
- 2.4. Identify stakeholders

3. Requirements Engineering

- 3.1. Gather requirements
- 3.2. Identify user requirements
- 3.3. Identify system requirements
- 3.4. Write functional requirements
- 3.5. Write non-functional requirements

3.6. Requirements analysis (Conflicts & Ambiguities)

4. System Modeling

- 4.1. Create Use Case Diagram
- 4.2. Create Context Diagram
- 4.3. Create Activity Diagram
- 4.4. Create Gantt Chart
- 4.5. Create PERT Diagram
- 4.6. Create State Machine Diagram (Bonus)
- 4.7. Explain models & connections

5. System Design / Prototype

- 5.1. Design user interface
- 5.2. Create database design
- 5.3. Develop web prototype (Bonus)

6. Testing

- 6.1. Check system functions
- 6.2. Fix errors
- 6.3. Improve usability

7. Final Deliverables

- 7.1. Prepare final report
- 7.2. Prepare diagrams sheet
- 7.3. Review project rubric sheet
- 7.4. Submit project

5 Requirements Engineering

5.1 User Requirements

User requirements describe what the users need from the system. They are written in simple language from the user's point of view.

- As a student, I want to log in with my university ID so I can access my account.
- As a student, I want to see all available courses so I can choose what to study.
- As a student, I want to search for a course by name or code so I can find it quickly.
- As a student, I want to register for courses online so I do not have to go to the office.
- As a student, I want the system to tell me if I am missing a prerequisite so I know what I need first.
- As a student, I want the system to stop me from joining a full class so I know the seat is available.
- As a student, I want to drop a course before the deadline if I change my mind.
- As a student, I want to see my class schedule clearly so I can plan my time.
- As a student, I want to get a confirmation after I register so I know it worked.
- As an admin, I want to add and manage courses each semester so the system stays up to date.
- As a faculty member, I want to see who is registered in my course so I can prepare my class.
- As a student, I want to see my past registrations so I can track my academic history.

5.2 System Requirements

System requirements describe how the system fulfills user needs at a technical level.

- The system shall authenticate users using their university ID and password with secure session management.
- The system shall retrieve and display all courses available for the active semester from the database.
- The system shall provide a search function that filters courses by name, code, or department in real time.
- The system shall process course registration requests and update the database to reflect the student's enrollment.
- The system shall query the prerequisites table and reject registration if the student has not completed required courses.
- The system shall check the current enrollment count against the course capacity limit and block registration if the course is full.
- The system shall allow students to submit a drop request and update enrollment records accordingly before the drop deadline.
- The system shall generate and display a weekly schedule view based on the student's current registered courses.
- The system shall send an automated email or in-system notification to the student after any successful registration or drop action.
- The system shall provide an admin interface to create, update, and delete course records including seat limits and schedule details.
- The system shall display a real-time list of enrolled students for each course, accessible only to the course's assigned faculty.
- The system shall maintain and display a history log of all past course registrations linked to the student's account.

5.3 Functional Requirements

Functional requirements describe what the system should do.

ID	Requirement	Description
FR-01	Student Login	The system must allow students to log in using their university ID and password.
FR-02	View Courses	The system must show students a list of all available courses for the current semester.
FR-03	Search for Courses	Students must be able to search for courses by name, code, or department.
FR-04	Register for Courses	The system must allow students to choose and register for courses they want to take.
FR-05	Check Prerequisites	Before a student registers, the system must check if they have completed all required prerequisite courses.
FR-06	Check Capacity	The system must check if a course still has available seats before it allows registration.
FR-07	Drop a Course	Students must be able to drop a course they are already registered in before the deadline.
FR-08	View Schedule	The system must show each student their personal class schedule after registration.
FR-09	Send Confirmation	After a student registers or drops a course, the system must send a confirmation message or email.
FR-10	Admin Manage Courses	The admin must be able to add, edit, or remove courses and set the maximum number of seats.
FR-11	Faculty View Students	Faculty members must be able to see the list of students registered in their courses.
FR-12	View Registration History	Students must be able to view their past course registration history.

5.4 Non-Functional Requirements

Non-functional requirements (NFRs) define the quality attributes and constraints the system must satisfy. Unlike functional requirements, they describe *how* the system performs rather than *what* it does.

ID	Category	Description
NFR-01	Performance	Registration requests must be processed within 5 seconds under normal load (up to 500 concurrent users).
NFR-02	Availability	The system must be available 99.5% of the time, with maintenance scheduled outside peak registration periods.
NFR-03	Security	All data must be transmitted over HTTPS. Passwords must be hashed and sessions must expire after 30 minutes of inactivity.
NFR-04	Usability	Key tasks like registration and schedule viewing must be completed in no more than 4 steps.
NFR-05	Scalability	The system must handle a 2x–3x increase in users during peak registration without performance degradation.
NFR-06	Reliability	No registration data should be lost on failure. The system must recover using automatic backups.
NFR-07	Maintainability	Code and database structure must be documented so a new developer can understand and modify any module.

6 Requirements Analysis

All functional and non-functional requirements were reviewed to identify unclear or conflicting points. Findings are split into two parts: ambiguities and conflicts.

6.1 Ambiguities Found

An ambiguity occurs when a requirement can be interpreted in more than one way, or when it omits information necessary to understand the expected behavior.

ID	Affected Req.	Ambiguity Found	Suggested Resolution
AMB-01	FR-07: Drop a Course	The “deadline” for dropping a course is never defined. It is unclear if it is a fixed date or varies per course.	Define a specific drop deadline per semester and display it to the student.
AMB-02	FR-09: Send Confirmation	The requirement says “message or email” without specifying which channel is primary.	Choose one primary channel (e.g., email only).
AMB-03	FR-11: Faculty View Students	The requirement does not specify which student data is visible to faculty, creating a potential privacy risk.	Define which fields are shown (e.g., name and ID only).
AMB-04	FR-10: Admin Manage Courses	It is unclear whether multiple admin accounts are allowed, and what happens if an admin deletes a course with enrolled students.	Define a single admin role and specify that deleting a course with active enrollments must be blocked.
AMB-05	FR-01: Student Login	The requirement does not define what happens after multiple failed login attempts. No lockout or brute-force protection is mentioned.	Define a maximum number of failed attempts (e.g., 5) after which the account is temporarily locked.

6.2 Conflicts Found

A conflict occurs when two or more requirements cannot both be fully satisfied simultaneously, or when satisfying one makes it harder or impossible to satisfy another.

ID	Between	Conflict Description	Resolution Approach
CON-01	FR-04 vs FR-06	FR-04 allows free registration while FR-06 blocks it if the course is full. Without a clear order of checks, both cannot be satisfied simultaneously.	Always check seat availability (FR-06) before confirming registration (FR-04). FR-06 takes priority.
CON-02	NFR-01 vs NFR-05	NFR-01 requires a 5-second response under 500 users, while NFR-05 requires handling 2x–3x more users. Satisfying both demands higher infrastructure costs.	Use load balancing and auto-scaling to meet both requirements.